Dataset for XSL

20160510
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SUN database for computer vision

- Scene “Street”
    - Image-01
        - Object “Person”
        - Object “Building”
        - Object “Car”
        - Object “Bike”
    - Image-02
        - Object “Person”
        - Object “Tree”
        - Object “Car”
        - Object “Car”
    - Image-03
        - Object “…”

Homepage: http://groups.csail.mit.edu/vision/SUN/

SUN database for computer vision

- Homepage:
  - http://groups.csail.mit.edu/vision/SUN/
- Image index page:
  - http://labelme.csail.mit.edu/Images/users/antonio/static_sun_database/
- Annotation index:
  - http://labelme.csail.mit.edu/Annotations/users/antonio/static_sun_database/
- MATLAB util code:
  - https://github.com/CSAILVision/LabelMeToolbox/tree/master/SUNdatabase

SUN database for computer vision

- 131067 Images
- 908 Scene categories
- 313884 Segmented objects
- 4479 Object categories

MEMO
- #images per scene category is not uniform
- #annotations per image is not uniform

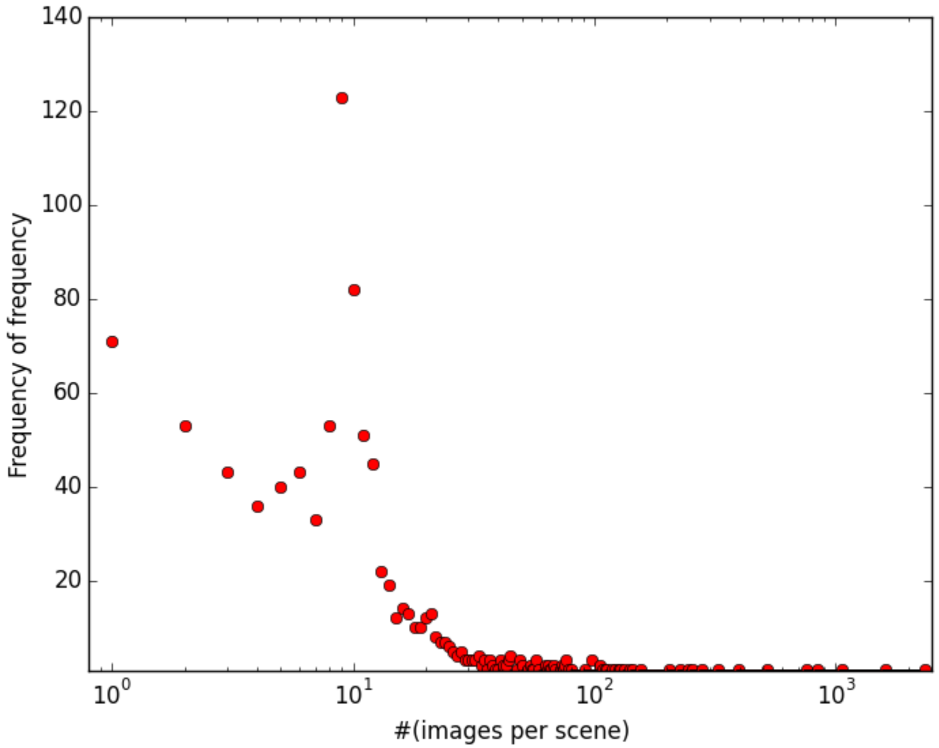
CAUTION: Hereafter I say
“object” instead of “object category”;
“scene” instead of “scene category”

MEMO
- Scenes have subcategories
  - indoor
  - outdoor
  - etc
- Objects have commom postfix
  - crop
- No annotation guideline?
  - “car”
    vs “cars”
    vs “two cars”
    vs “there are cars”

Fix and omit name patterns

- Omit: Name, #words > 3
  - E.g. “Bunch of white floweres”
- Omit: Irregular annotations (“is” or “are” or “.”)
  - E.g. “There is a car.”
- Fix: Name, ends with “crop”
  - E.g. “Person crop” → “Person”
- Fix: Plural → Singular
  - E.g. “Cars” → “Car”

#(images per scene)



#(images per scene) (top 10)

street 2336

bedroom 1603

living_room 1062

kitchen 839

bathroom 759

dining_room 522

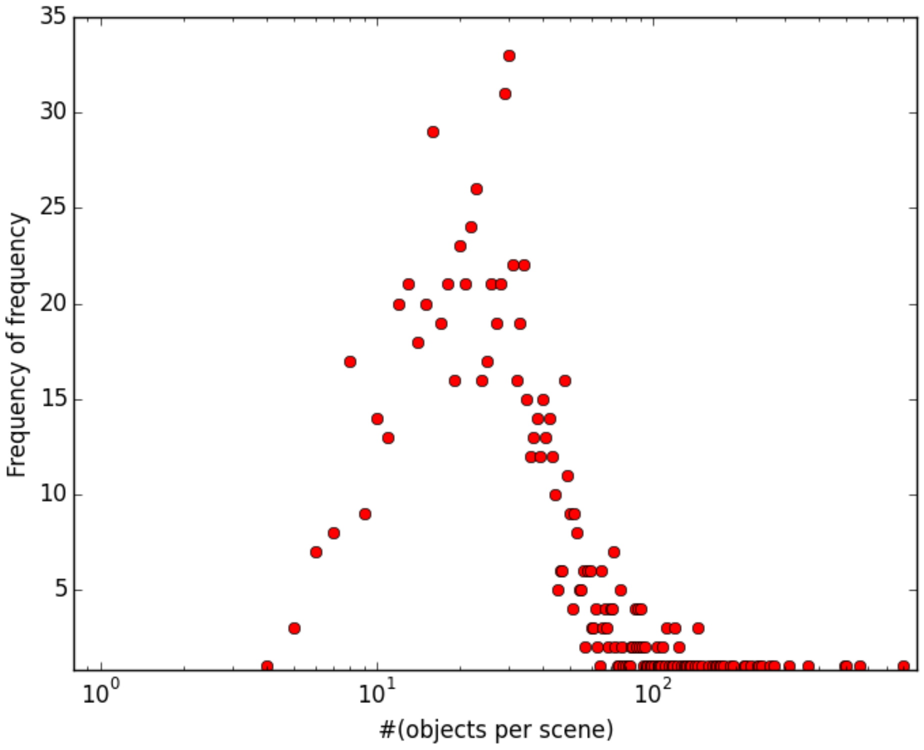
skyscraper 397

highway 326

mountain_snowy 278

building_facade 253

#(objects per scene)



#(objects per scene) (top 10)

street 805

kitchen 558

living_room 501

bedroom 495

dining_room 363

bathroom 311

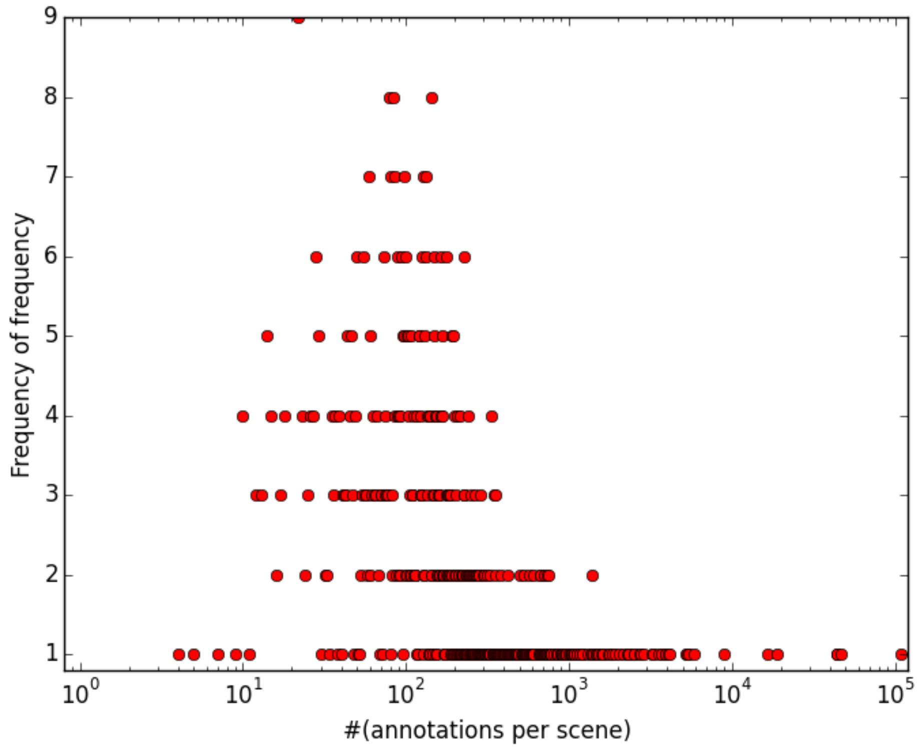
home_office 274

conference_room 266

hotel_room 247

office 242

#(annotations per scene)



#(annotations per scene) (top 10)

street 108414

living_room 46449

bedroom 44507

kitchen 43870

dining_room 19096

bathroom 16686

conference_room 8924

building_facade 5817

highway 5430

skyscraper 5312

Scene-object occurrence matrix

Scene-object matrix

	Person	Bldng	Car	Bike	Tree	…
Street	2	1	3	1	1	
Kitchen	2	0	0	0	1	
…						

- Scene “Street”
  - Image-01
    - Object “Person”
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    - Object “Car”
    - Object “Bike”
  - Image-02
    - Object “Person”
    - Object “Tree”
    - Object “Car”
    - Object “Car”

4 annotations
4 objects

4 annotations
3 objects

2 images
5 objects

- “Person” x 2
- “Building” x 1
- “Car” x 3
- “Bike” x 1
- “Tree” x 1

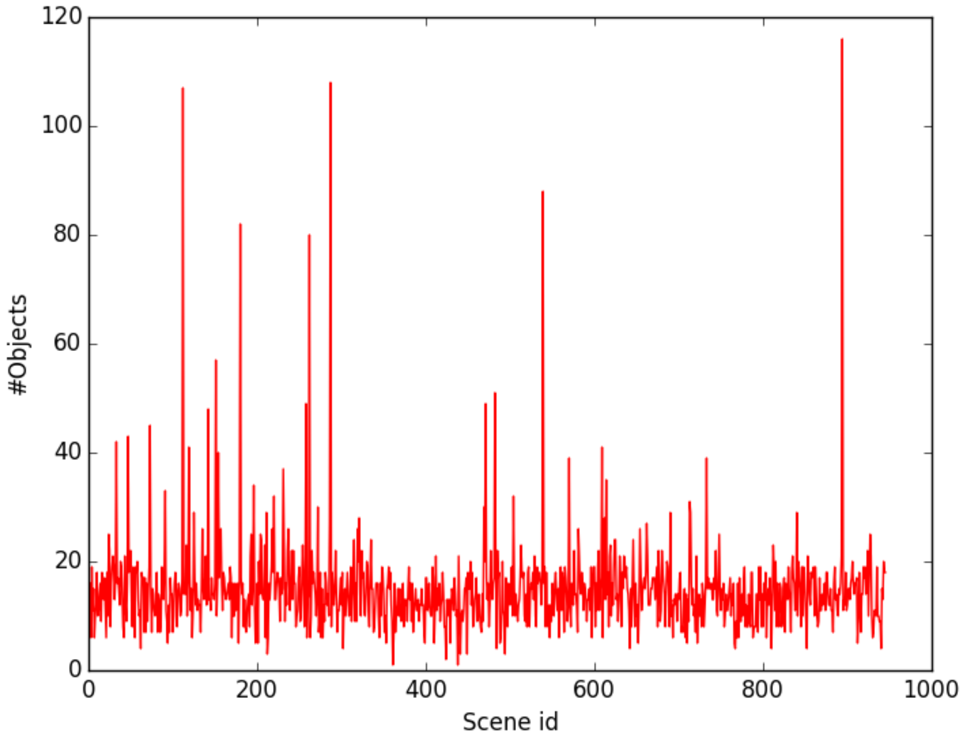
Scene-object matrix, size reduction

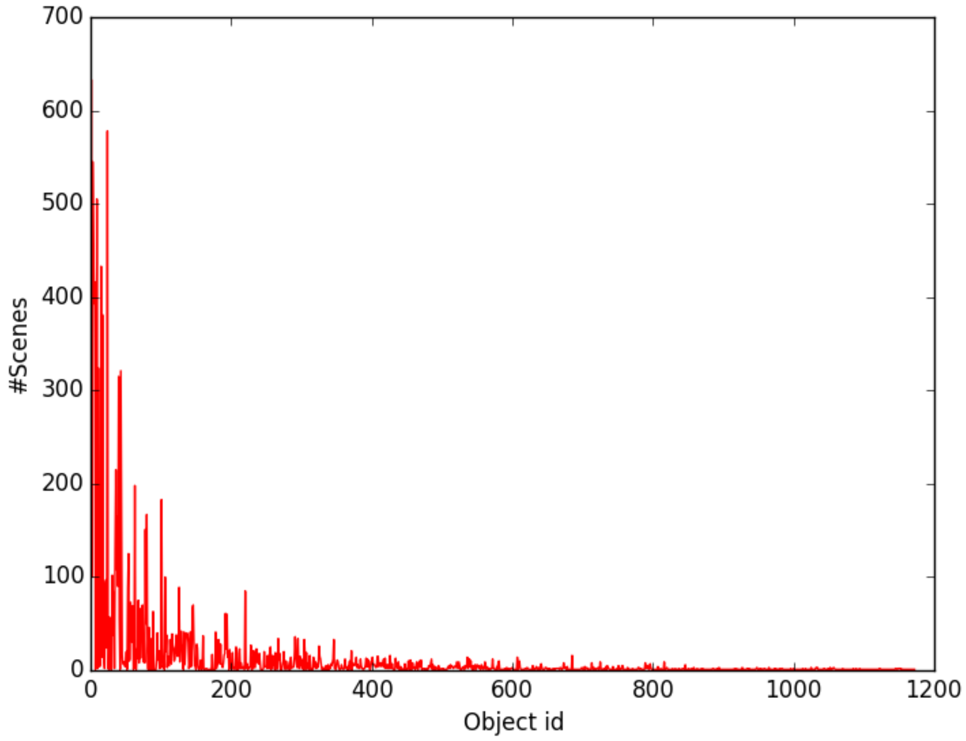
- Scene-object matrix of size 1000 x 4500
  - Too large
- Non-uniform #images per scene

- Size reduction
  - Keep object-categories commonly appear in alpha% of images
    - #objects > #images × θ
    - Set $\theta = 0.05$
  - Size 942 x 1172
- (i, j)-element
  - #images the object j appear in the scene i

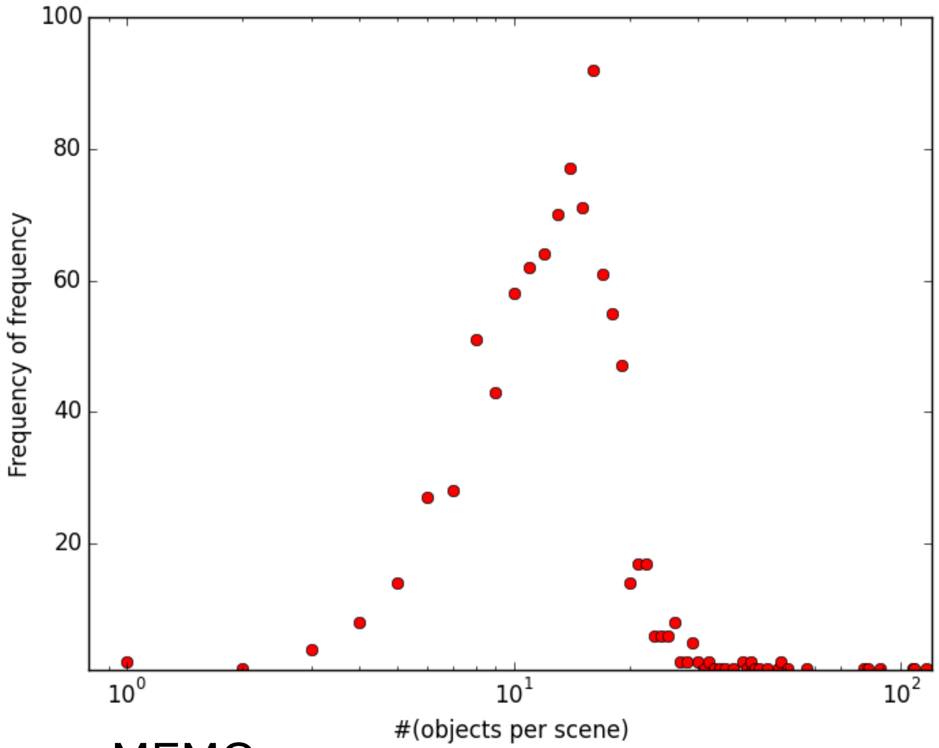
#(objects per scene), #(scenes per object)

MEMO:
Since assigning id *in discovery order(?)*,
frequent objects tend to have smaller ids





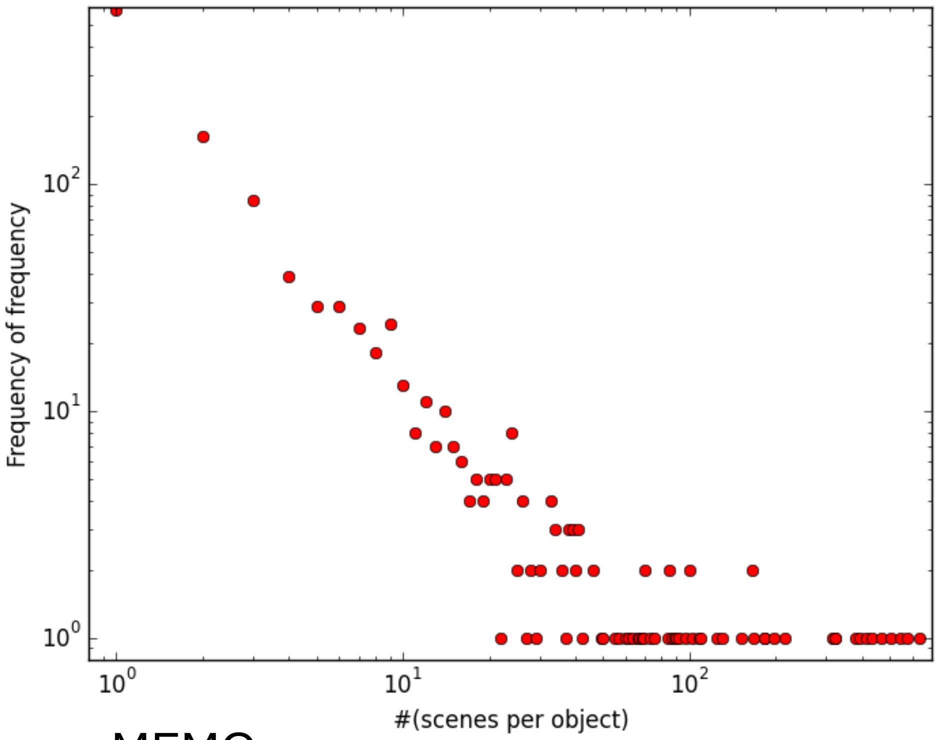
#(objects per scene)



MEMO:
After preprocessing (fix & omit),
Some scenes has only one object

#(objects per scene) (top 10)
kitchen 116
bedroom 108
living_room 107
street 88
dining_room 82
bathroom 80
hotel_room 57
parlor 51
poolroom/home 49
building_facade 49

#(scenes per object)



MEMO:
Lots of objects in SUN database appears only one scene

#(scenes per object) (top 10)

wall 633

person 578

sky 545

tree 505

window 465

floor 433

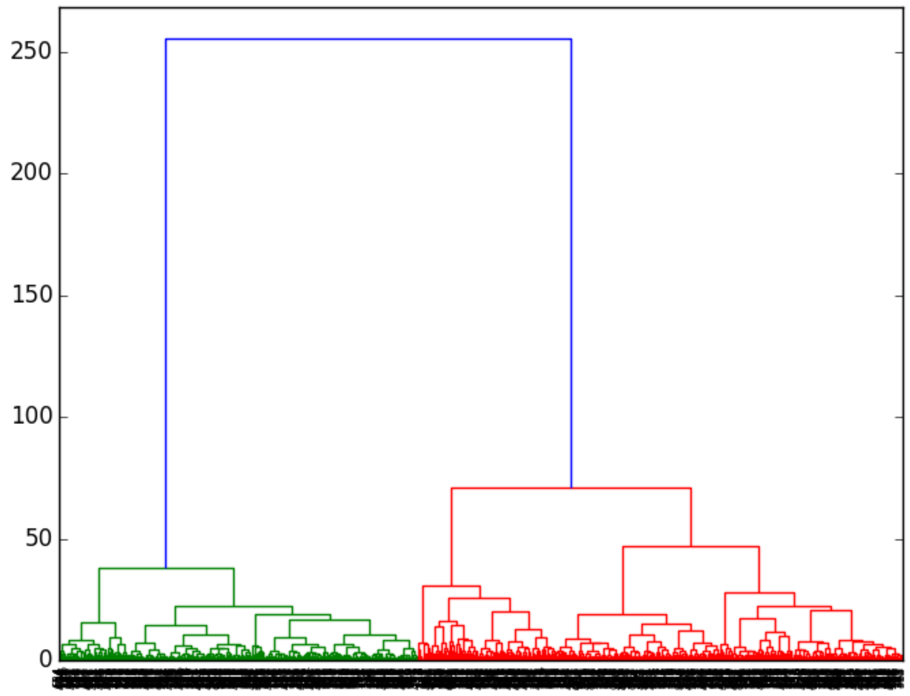
door 416

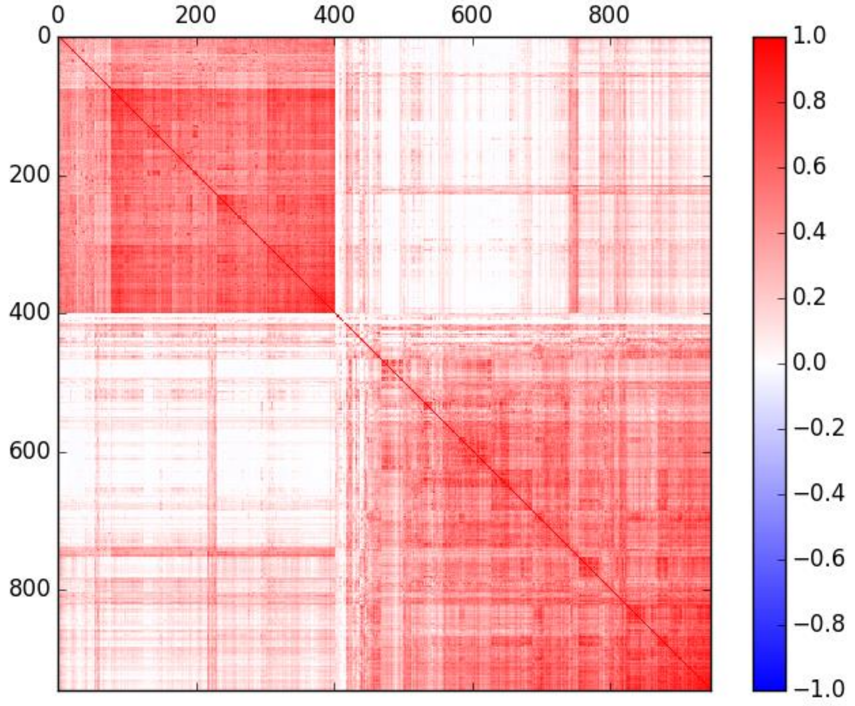
building 393

ceiling 381

grass 324

Correlations among scenes (w/ object vectors)

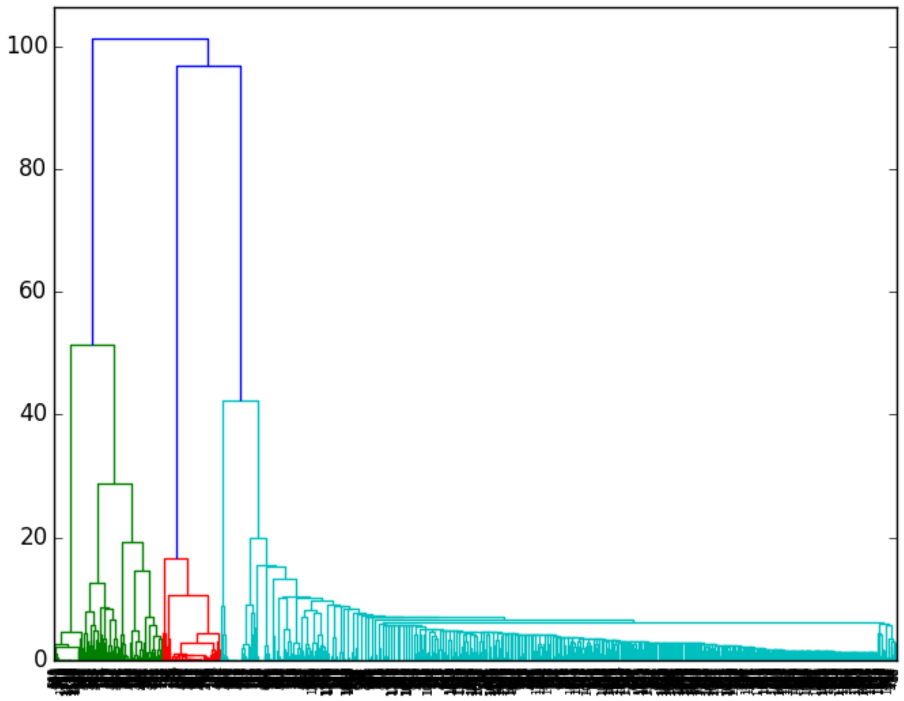




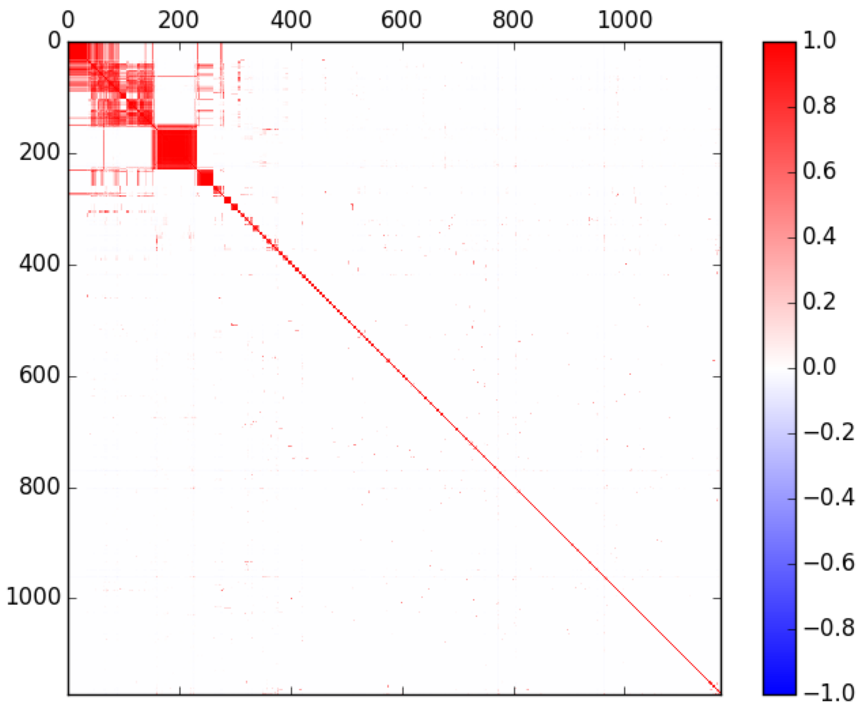
indoor outdoor

indoor outdoor

Correlations among objects (w/ scene vectors)



kitchen stuffs
bedroom stuffs
else?

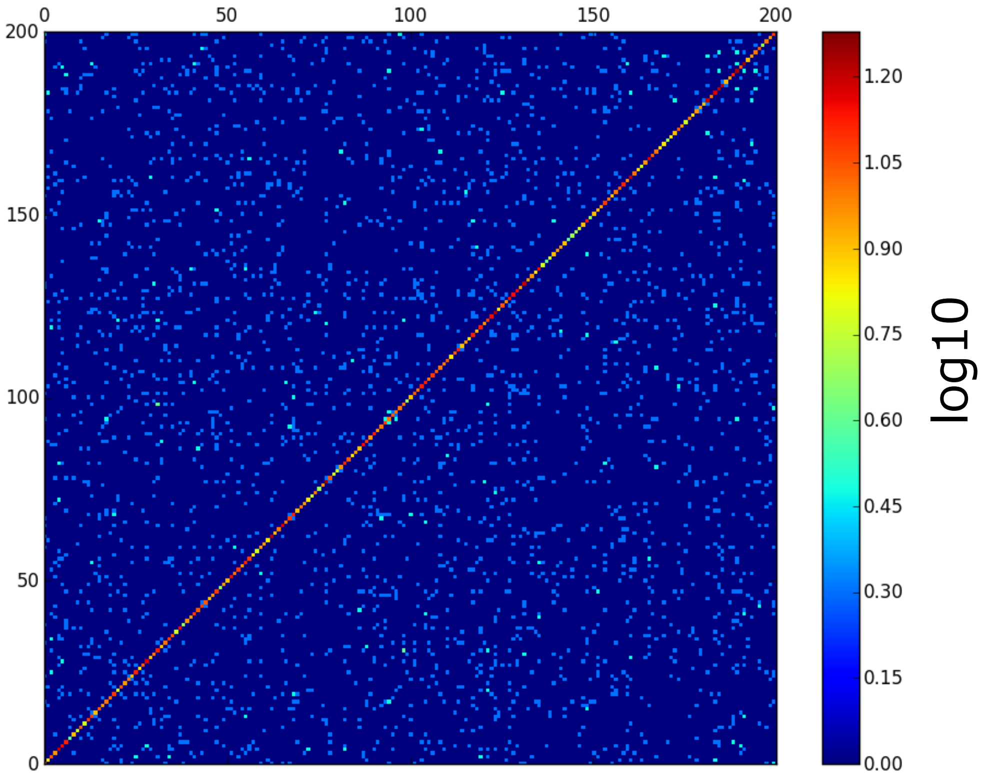


XSL simulation with SUN database

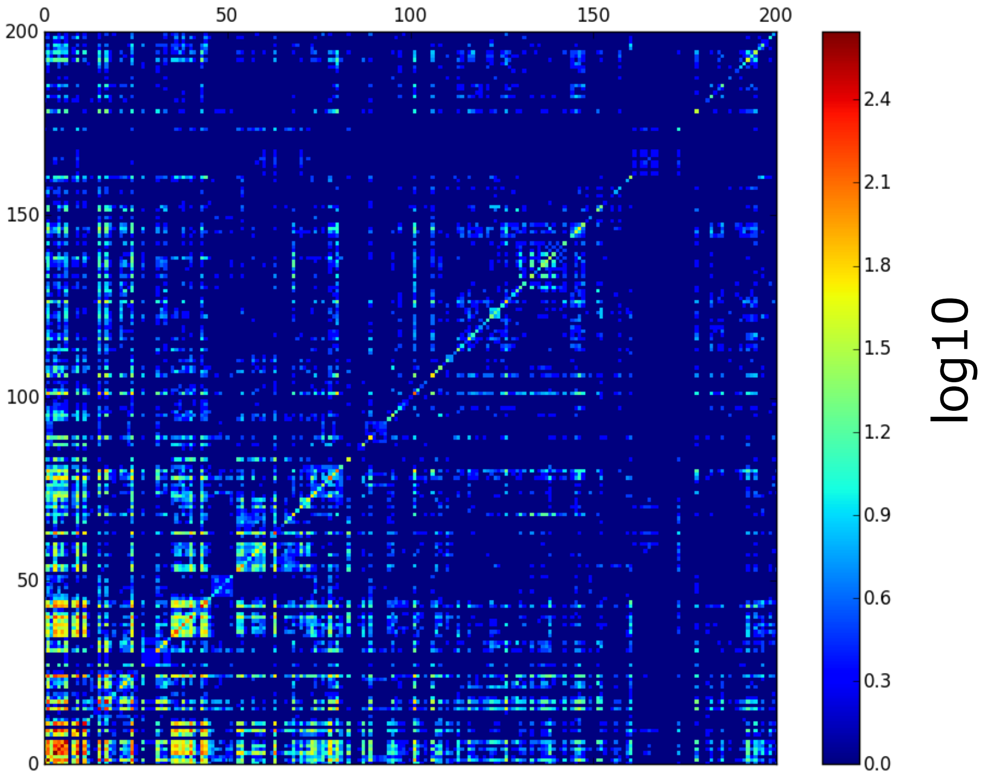
Procedure

- From the scene-object matrix,
  - Choose at random a scene
  - Choose M objects, composed of the scene
  - Choose N words (from M objects if $N \leqq M$)
- Expose N words, M objects
  - Always including the correct pairs
- Learn all the N x M possibilities
  - Counting their frequency

Uniform choice (1072 word-object pairs)



SUN database (1072 word-object pairs)



Zipf distribution (1072 word-object pairs)